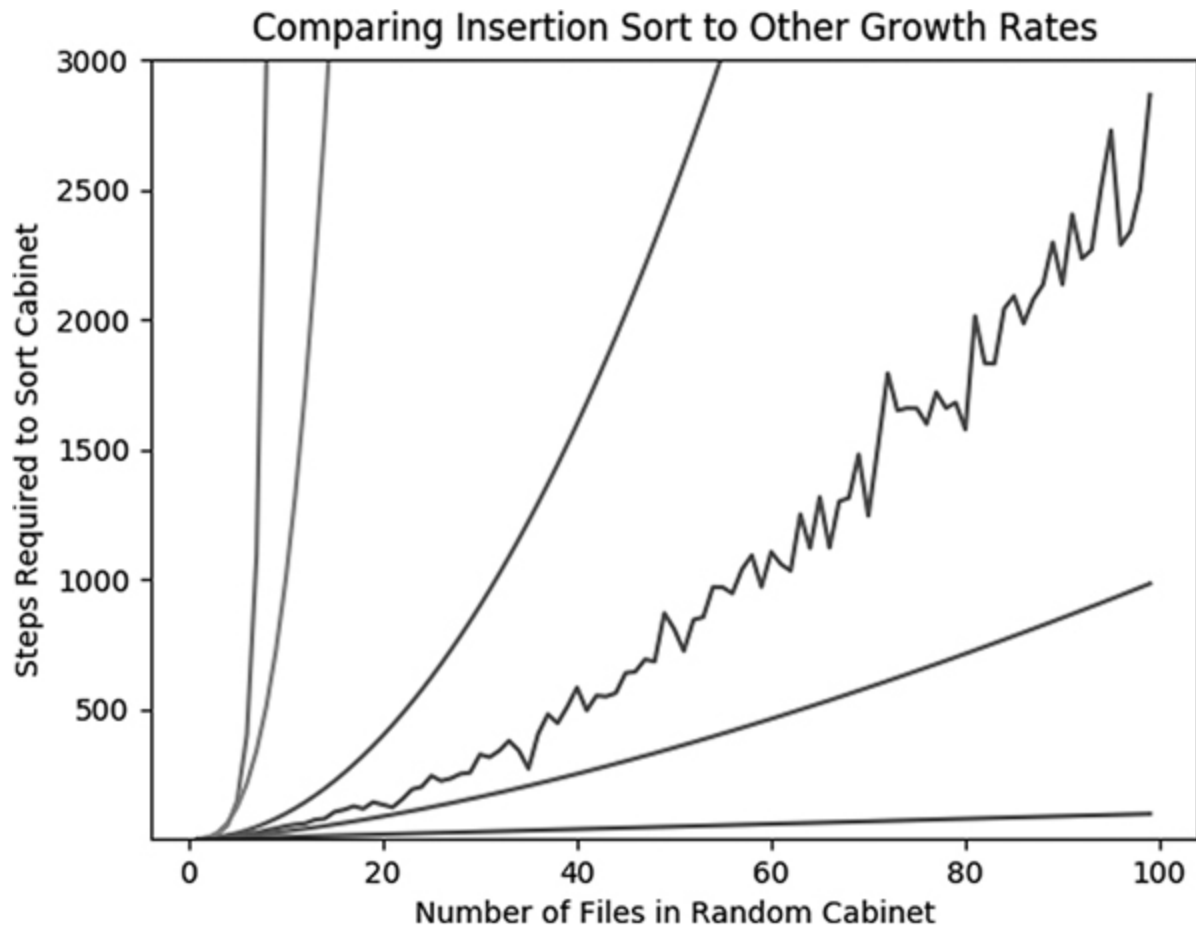




[Figure 4-3](#): Insertion sort steps compared to other growth rates

There are five growth rates plotted in [Figure 4-3](#), in addition to the jagged curve counting the steps required for insertion sort. You can see that the exponential curve grows the fastest, and next to it the cubic curve scarcely even makes an appearance on the plot because it also grows so fast. The $y = x$ curve grows extremely slowly compared to the other curves; you can see it at the very bottom of the plot.

The curves that are the closest to the insertion sort curve are $y = x^2$ and $y = x^{1.5}$. It isn't obvious which curve is most comparable to the insertion sort curve, so we cannot speak with certainty about the exact growth rate of insertion sort. But after plotting, we're able to make a statement like "if we are sorting a list with n elements, insertion sort will take somewhere between $n^{1.5}$ and n^2 steps." This is a more precise and robust statement than "as long as a gnat's wing flap" or ".002-ish seconds on my unique laptop this morning."